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Label-Flipping Poisoning Attack Against ML-based NIDS: A Comparative Analysis

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***Abstract*— Machine learning (ML) and deep learning (DL) techniques are aggressively adopted by modern network intrusion detection systems (NIDS) due to their ability to identify novel, complex, and sophisticated attack patterns. However, the reliability of these systems is highly dependent on the quality of the training dataset. In this work, we investigated the vulnerability of ML and DL based NIDS against training time poisoning attack by using the label flipping approach. We used CIC-IDS2017 dataset, and conducted a comparative analysis of seven widely used classifiers including, Decision Tree, Random Forest, Logistic Regression, Gaussian Naive Bayes, Linear Support Vector Classifier, XGBoost, and a Deep Neural Network (DNN). We performed our experiment by formulating binary classifier and evaluated the degradation in learning behavior with increasing percentage of poisoning. Our experimental results shows that most of the ML models are vulnerable to poisoning attack, we seen a linear degradation with progressive poisoning percentage, whereas models like Decision Tree, and Random forest has shown resistance against these attacks. We explicitly evaluated our framework on training data itself, to understand the learning behavior of various models under poisoning attack. At 50\% poisoning, the accuracy of DNN for attack class has dropped from 99.25\% to 12.61\%, while the tree-based models maintained accuracy above 99\%. Our findings reveal a critical vulnerability of ML and DL-based NIDS, and emphasize the need of robust training strategies against poisoning attacks.**

Keywords*— *Network Intrusion Detection System, Adversarial Attacks, Poisoning Attack, Label Flipping, Machine Learning, Deep Learning, Security

I. Introduction

Network Intrusion Detection Systems (NIDS) are a critical component of modern cybersecurity infrastructures; they continuously monitor the network traffic to detect malicious activities. Anomaly-based NIDS are being developed to overcome the limitations of signature-based systems in detecting zero-day attacks by using machine learning (ML) and deep learning (DL) techniques to model normal traffic patterns; these systems mark any deviation as a potential threat [1], [2]. The effectiveness of such systems is highly dependent on the quality and integrity of the training data [3]. Benchmark datasets such as CIC-IDS2017 have shown high detection accuracy for ML-based models [4]. Despite their advantages, ML/DL-based NIDS introduces new security challenges in the form of adversarial attacks. These attacks are broadly categorized into two categories: training-time poisoning attacks and test-time evasion attacks. In poisoning attacks, the adversary manipulates the training data to corrupt the model's learning process with the goal of degrading the performance of the model. Label flipping is one of the simple but effective strategies in which the labels are intentionally altered to mislead the learning process [5]. Prior studies have shown that even small alterations in training data can significantly degrade the performance of DL-based NIDS [6]. However, limited attention has been paid to systematic comparisons of classical ML models and DL models under poisoning attacks, especially for binary anomaly detection scenarios [7].

Motivated by this gap, this paper presents a comparative analysis of label-flipping poisoning attacks against seven widely used classifiers on the CIC-IDS2017 dataset. We formulate a binary intrusion detection problem and analyze the degradation trends under increasing poisoning rates. We provided a unified comparison across various classifiers including Decision Tree, Random Forest, Logistic Regression, Gaussian Naive Bayes, Linear Support Vector Classifier, XGBoost, and a Deep Neural Network.

The remainder of this paper includes Related Works, Preliminaries, Experimental Setup, Proposed Methodology, Experimental Results, and Conclusion.

II. Related Works

Various studies have been carried out recently on adversarial attacks against ML-based security systems. Apruzzese et al. [1] provided an overview of realistic threats against malware detection, spam filtering, and IDS. They highlighted a practical impact of poisoning and evasion attacks on deployed defense systems. Praveena et. al. [5] categorized adversarial attacks on the basis of the attacker's goals, knowledge, and capabilities. They identified the poisoning attacks as an unresolved issue for the systems that rely on automated data collection pipelines.

Mohammadian et al. [6] evaluated label flipping poisoning attacks against DL-based NIDS, which is trained on CIC-IDS2017, and revealed that DL-based NIDS are highly vulnerable under poisoning attacks. They extended their research in [7] by jointly analyzing the combined effect of poisoning and evasion attacks, showing that the impact of adversarial attacks significantly varies across different

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datasets and attack configurations. Similar vulnerabilities have been demonstrated for classical ML models in [8], where they show the effect of poisoning and evasion attacks in logistic regression and other linear classifiers.

Some defense strategies, such as data sanitization, adversarial retraining, robust loss functions, and ensemble methods, have been explored [9]. However, these strategies can introduce computational overhead and reduce the performance of models in the absence of adversarial attacks. Biggio and Roli [10] emphasized that poisoning attacks are highly concerning for continuously learning systems.

Early studies have shown the vulnerability of SVMs to training-time poisoning attacks [11], while training with noisy data has been formally analyzed in [12]. Recent empirical studies, such as [13], have demonstrated that traditional ML models show different sensitivity to poisoning attacks across various IDS datasets. Their study highlights the necessity for systematic evolution for robustness. Ensemble techniques like random forest [14] and gradient boosting methods such as XGBoost [15] are commonly adopted in classification tasks, which motivates their inclusion in comparative poisoning studies.

Although prior studies have demonstrated the effectiveness of label flipping poisoning attacks and explored adversarial threats in NIDS, there is still a gap in conducting a systematic comparison of a broad range of classical ML and DL models under similar preprocessing and poisoning conditions. Our work addressed this gap, and we conducted a comparative analysis of seven widely used classifiers trained on the CICIDS2017 dataset, which provides a deep insight into the robustness of various models under label flipping poisoning attacks.

III. Preliminaries

A. Network Intrusion Detection Systems

A Network Intrusion Detection System (NIDS) continuously monitors network traffic to detect any ongoing anomalous or malicious activities on the network. There are two approaches to developing NIDS: signaturebased NIDS and anomaly-based NIDS. Signature-based NIDS maintains a database of known signatures and matches the traffic flow with existing stored signatures. As these systems rely on attack signatures, they are only capable of detecting known attacks and fail to detect novel (never-seen-before) attack patterns. To overcome this problem, machine learning techniques are used to develop anomaly-based NIDS, which treats intrusion detection as a classification problem where traffic flows are labeled as benign or an attack. Anomaly-based NIDS draws a baseline of normal network flow and marks any deviation as a potential attack.

B. Dataset

We used the CIC-IDS2017 dataset for our experiment, which is a benchmark dataset created by the Canadian Institute of Cybersecurity [4]. This dataset represents real-world network traffic by containing a large amount of benign and attack data. The dataset contains 2273097 benign and 558646 attack records of various attack classes, including DDoS, PortScan, Botnet, Infiltration, Web Attack-Brute Force, and Web Attack-Sql injection, Web Attack-XSS, FTP-Patator, SSH-Patator, DoS GoldenEye, DoS Hulk, DoS Slowhttp, DoS Slowloris, and Heartbleed.

C. Label Flipping Poisoning Attack

Label flipping is an approach to mount the poisoning attack in which the attacker modifies the class labels of selected training samples. In binary classification, labels are flipped from benign to the attack class and vice versa. The goal of this attack is to corrupt the learning process of the ML model, which results in learning incorrect decision boundaries and degrades the detection performance.

D. Models Evaluated

- Classical ML models: DecisionTreeClassifier, RandomForestClassifier, LogisticRegression, GaussianNB, LinearSVC, and XGBClassifier.
- Deep neural network (DNN): Sequential model with 68 inputs, two hidden layers with 256 neurons with ReLU activation function followed by Dropout layer with 0.2 probability, and an output layer with two neurons and softmax activation function. We used a sparse categorical crossentropy loss function and an Adam optimizer. The rest of the hyperparameters are default.

IV. Proposed Methodology

This section presents our proposed experimental methodology, including data preprocessing and label flipping attack pipelines.

A. Data Preprocessing

As mentioned in III-B, we used the CIC-IDS2017 dataset for our experiment. Initially, when we acquired the dataset, there were a total of 2830743 records, 78 features, and a label column in the dataset. We started by removing duplicate, zero-variance, and irrelevant features from the dataset, which reduced the dimensionality from 78 to 68 features. Then we removed the records containing duplicate, null, and infinite values. After this data cleaning process, we performed label encoding. We have given binary labels to the dataset; we marked

Benign records with 0 and the rest of the records with 1. The dataset was highly imbalanced, initially there were 2273097 Benign and 558646 attack records which makes it a ratio of 80:20. After performing data cleaning there were 1896672 benign and 336492 records left, and the ratio became 85:15 which makes the dataset highly imbalanced and is not suitable for model training. To overcome this issue, we used a data undersampling method to reduce the number of benign samples to match the number of attack samples. We performed simple random sampling to reduce the number of benign samples. We balanced the dataset in a 50:50 ratio. After balancing the dataset, we performed feature scaling using the standard scaler because most of these models require scaled records for better convergence. Figure 1 shows the

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graphical presentation of the whole data preprocessing pipeline.

B. Label-Flipping Poisoning Attack

We used a label flipping approach to mount the poisoning attack, which is a training-time adversarial attack in which the attacker manipulates labels of a subset of training data with the intention of corrupting the learning process. The feature values remain intact; there is no alteration performed in features, only the labels are altered to another class. In our case, first we select a subset from the training dataset based on the required fraction, and flip the labels from attack to benign and benign to attack. We use stratified sampling to select the subset, which keeps the equal representation of both classes. Given a poisoning rate ρ%, and number of total data in training set N the number of samples to be flipped Nflip is calculated as:

$$N_{flip} = \frac{\rho}{100} * N \quad (1)$$

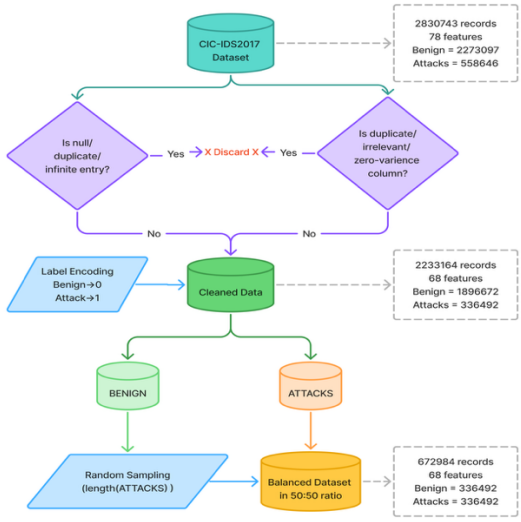


Fig.1 Data Preprocessing Pipeline

N_{flip} samples are selected from the original training dataset using stratified sampling. For each selected sample, the label is inverted using the following equation:

$$y_i' = 1 - y_i \quad (2)$$

This binary flipping strategy ensures that the benign samples are mislabeled as attacks and vice versa. After mislabelling, the poisoning dataset is constructed by replacing the original labels of the selected samples with the poisoned ones. We shuffled the poisoned dataset before training the models to eliminate bias.

We used poisoning rates of 0% (Clean data), 10%, 20%, 30%, 40%, and 50% for the evaluation to analyze the robustness of models under increasing attack intensities.

C. Model Training

We evaluated various machine learning and deep learning models to study the robustness of different learning paradigms. The models we evaluated are as follows:

- Decision Tree Classifier (DT)
- Random Forest Classifier (RF)
- Logistic Regression (LR)
- Gaussian Naive Bayes (GNB)
- Support Vector Classifier (SVC)
- Extreme Gradient Boosting (XGBoost)
- Deep Neural Network (DNN)

Each model is trained with poisoned training data with a different poisoning rate. Figure 2 shows our proposed attack framework.

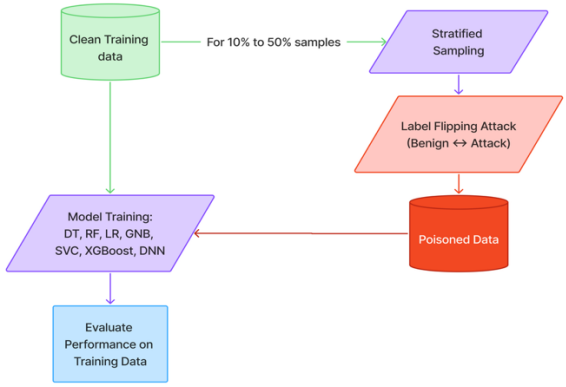


Fig.2 Proposed Label Flipping Attack Framework

Fig. 1. *DT: Decision Tree, RF: Random Forest, LR: Logistic Regression, GNB: Gaussian Naïve Bayes, SCV: Support Vector Classifier, XG Boost: Extreme Gradient Boosting, DNN: Deep Neural Network*

V. EXPERIMENTAL SETUP

We assumed a white box setting, in which the attacker has complete knowledge of the learning process and complete access to the training dataset. The attacker is capable of manipulating the labels under this setting. All experiments are conducted on a workstation running Windows 11 (64-bit), equipped with an Intel Core i5-1155G7 processor and 16 GB of RAM. No GPU acceleration is used, and all computations are performed on the CPU. We used the Python programming language and its libraries for implementation. For classical ML models, we used the scikit-learn library. A deep learning model is implemented using TensorFlow with the Keras API.

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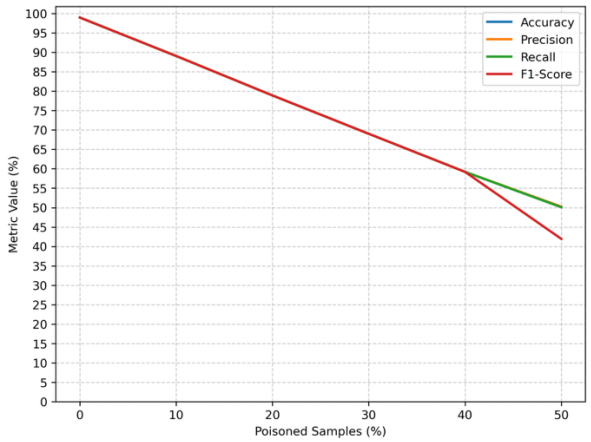
VI. RESULTS AND DISCUSSIONS

In this section, we analyzed the experimental results and compared the robustness of various models. Table I presents the results of the performance degradation of the DNN model. We can notice a linear degradation in the overall performance. When our DNN model is trained on clean data, the overall accuracy was 98.96%, and attack accuracy was 99.25%, but when 50% of the labels have been poisoned, the accuracy has decreased to 50.11% and 12.61%, respectively, for the overall and attack classes. The linear degradation of performance for detecting overall classes can be seen in Figure 3a, and the performance degradation of DNN to detect attack classes can be seen in Figure3b. The DNN model has been collapsed when 50% of the samples are poisoned and the model is no longer able to detect attack class.

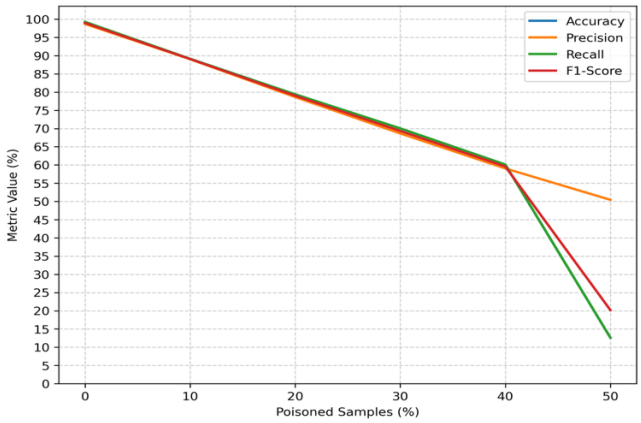
TABLE I. IMPACT OF LABEL-FLIPPING POISONING ATTACK ON DNN PERFORMANCE

Overall Performance (%)				
Poisoning	***Acc.***	***Prec.***	***Recall***	***F1***
0	98.96	98.97	98.96	98.96
10	89.06	89.06	89.06	89.06
20	78.90	78.90	78.90	78.90
30	69.02	69.03	69.02	69.02
40	59.21	59.27	59.21	59.21
50	50.11	50.33	50.11	41.94
Attack Class Performance (%)				
Poisoning	***Acc.***	***Prec.***	***Recall***	***F1***
0	99.25	98.68	99.25	98.97
10	89.13	89.01	89.13	89.07
20	79.38	78.62	79.38	79.00
30	70.08	68.64	70.08	69.33
40	60.12	59.05	60.12	59.58
50	12.61	50.43	12.61	20.17

As we performed this attack against various machine learning models, Table II shows a comparative analysis of the performance degradation of various models. Figure 4 visually represents the effect of poisoning attack using a heatmap of classifier accuracy across all poisoning levels. Figure 5 represents the overall performance degradation trends using line plots. We can clearly see that decision tree and random forest models have shown a high resistance against poisoning attack, models like XGBoost, Logistic Regression, Support Vector Classifier, and Deep Neural Network has shown a linear degradation, while Gaussian Na¨ıve Bayes has shown an unusual pattern, the performance has increased when trained with 20% poisoned and then the accuracy gradually dropped linearly.



a. *Overall performance degradation under label flipping attack*



b. Performance degradation of DNN for attack classes under label flipping attack

Fig. 3 Performance of DNN under label-flipping poisoning attack

TABLE II. IMPACT OF LABEL-FLIPPING POISONING ATTACK ON DNN PERFORMANCE

Pois (%)	**DT**	**RF**	**XGB**	**LR**	**SVC**	**GNB**	**DNN**
0	99.99	99.97	99.95	95.42	94.89	58.98	98.96
10	99.95	99.08	89.90	84.48	84.09	53.67	89.06
20	99.92	98.65	79.95	75.65	75.53	69.17	78.90
30	99.90	98.81	70.05	67.05	66.99	64.00	69.02
40	99.89	99.32	60.80	58.50	58.48	56.91	59.21
50	99.88	99.76	55.60	50.24	50.32	50.00	50.11

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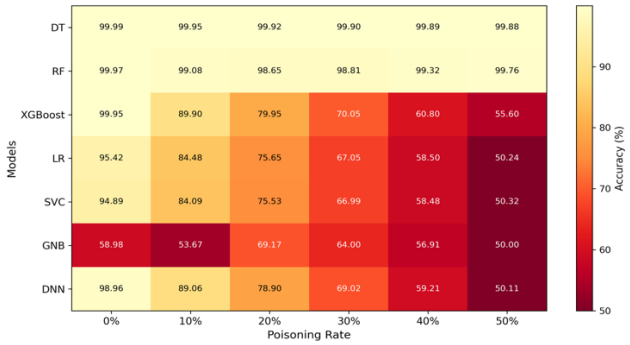


Fig. 4 Heatmap of overall accuracy (%) of different classifiers under varying label-flipping poisoning rates

Figure 6 shows the comparison of models trained with 50% poisoned samples,shows that even after poisoning 50% of the data, decision tree and random forest models remains resistant whereas other models face high degradation in performance. Even though decision trees and random forests have shown resistance, it's not guaranteed that the model has learned correct patterns, and it will perform as well as when trained with clean data for detecting unseen attack patterns. Such robustness of the model could be misleading, as it may indicate memorization rather than reliable learning of attack patterns. The results obtained have confirmed that machine learning and deep learning models are highly vulnerable to poisoning attacks.

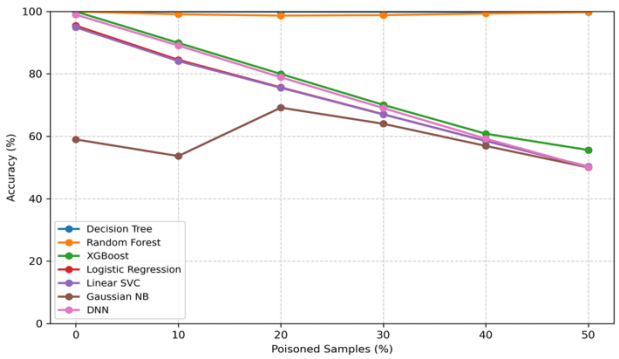


Fig. 5 Overall accuracy degradation of various ML and DL models under label-flipping poisoning attacks

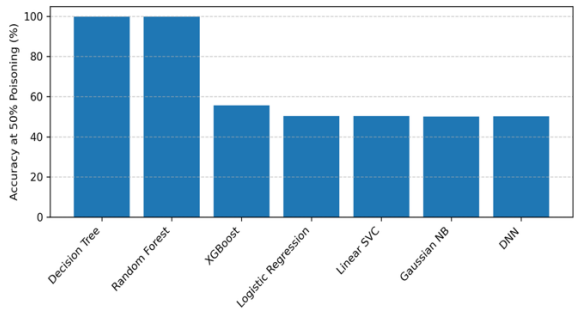


Fig. 6 Model Robustness at 50% Poisoning

VII. Conclusion

This paper presented a study of training time poisoning attack using label flipping approach against machine learning and deep learning based NIDS. we systematically evaluated the robustness of widely adopted classifiers including Decision Tree, Random Forest, Logistic Regression, Gaussian Na¨ıve Bayes, Linear Support Vector Classifier, XGBoost, and a Deep Neural Network under different percentage of poisoning rates. The experimental results demonstrates that most of the ML and DL models suffer from a significant degradation in performance as the poisoning rate increases, these models have been found near-linear decline in detection accuracy. Particularly the deep neural network has been completely collapsed for detecting attack class, initially the accuracy was 99.25% on clean data, declined to 12.61% when 50% labels has been flipped. This highlights a severe vulnerability of DNN against training data manipulation attacks. This study confirms that training time poisoning attacks are a serious threat for anomaly-based NIDS and the robustness of the system highly depends upon the quality of the dataset and learning paradigms. Future work can be done to investigate and develop defense mechanisms to protect ML/DL models from such attack vulnerabilities.

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